[ ]:
```
while loop syntax

intilization
while(condition):
    statements
    inc/dec
```

[9]:
```
i=1
while(i<=5):
    print(i)
    i=i+1
```

```
1
2
3
4
5
```

[11]:
```
#odd nos between 1 to 20
i=1
while(i <= 20):
    print(i)
    i=i+2
```

```
1
3
5
7
9
11
13
15
17
19
```

[10]:
```
#odd nos between 1 to 20
i=1
while(i <= 20):
    if(i%2!=0):
        print(i)
    i=i+1
```

```
1
3
5
7
9
11
13
15
17
19
```

[13]:
```
'''a=10
   b=20
   swap
   a=20
   b=10'''

a = 10
b = 20
print(f"before swap : \na ={a},b={b}")
temp=a
a=b
b=temp
print(f"after swap : \na ={a},b={b}")
```

```
before swap :
a =10,b=20
after swap :
a =20,b=10
```

[14]:
```
a = 10
b = 20
print(f"before swap : \na ={a},b={b}")
a,b = b,a
print(f"after swap : \na ={a},b={b}")
```

```
before swap :
a =10,b=20
after swap :
a =20,b=10
```

[ ]:
```
#fibonacci series
a=0
b=1
i=1
while(i<=8):
    print(a)
    sum = a+b
    a,b = b,sum
    i += 1
```

[ ]:
```
#fibonacci series
a=0
b=1
i=1
while(i<=8):
    print(a)
    sum = a+b
    a,b = b,sum
    i += 1
```

[7]:
```
#241 = 2+4+1=7
original = int(input("Enter a number"))
sum =0
while(original > 0):
    digit = original%10
    sum =sum+digit
    original//=10
print(sum)
```

```
Enter a number 472
13
```

[15]:
```
num =  int(input("Enter a number"))
original = num
reverse=0
while(original > 0):
    rem = original%10
    reverse = reverse*10+rem
    original//=10
print("number after reverse",reverse)
if(num == reverse):
    print("palindrome number")
else:
    print("not a palindrome number")
```

```
Enter a number 123
number after reverse 321
not a palindrome number
```

[23]:
```
'''* * * * *
   * * * * *
   * * * * *
   * * * * *
   * * * * * '''

for i in range(0,5):
    for j in range(0,5):
        print("*",end=" ")
    print()'''* * * * *
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

[2]:
```
'''
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
'''

for i in range(1,8):
    for j in range(1,6):
        print(j,end=" ")
    print()
```

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```